Exercise 7.8.4. Show that any exact sequence of vector spaces is isomorphic to a direct sum of complexes of the form

$$0 \to V \to V \to 0,$$

where V stands at the places i and $i+1$ and the map $V \to V$ is the identity (in particular, any short exact sequence of vector spaces is split). Is this true in the category of abelian groups?

Problem 7.8.5. Let $D_\bullet$ be a complex of abelian groups with differentials d_i, $i \in \mathbb{Z}$, let $C_\bullet$ be a subcomplex of $D_\bullet$ (i.e. a collection of subgroups $C_i \subset D_i$ such that $d_i(C_i) \subset C_{i+1}$), and let $E_\bullet = D_\bullet/C_\bullet$ be the quotient complex (i.e., $E_i = D_i/C_i$ with differentials induced by d_i).

Define a homomorphism $c_i : H^i(E) \to H^{i+1}(C)$ as follows. Given $x \in H^i(E)$, pick a representative x' of x in E_i. Let x'' be a lift of x' to D_i. Let $y' = dx'' \in D_{i+1}$ (we abbreviate the notation, denoting d_i just by d). Since $dx' = 0$, $y' = dx'' \in C_{i+1}$. Also, $dy' = d^2x'' = 0$. So y' represents an element $y \in H^{i+1}(C)$. We will set $c_i(x) = y$.

(i) Show that c_i is well defined, i.e., $c_i(x)$ does not depend on the choice of x' and x''.

(ii) Show that the sequence

$$\ldots H^{i-1}(E) \to H^i(C) \to H^i(D) \to H^i(E) \to H^{i+1}(C) \ldots, \tag{7.8.1}$$

where the first map is c_{i-1}, the middle two maps are induced by the maps $C_i \to D_i \to E_i$, and the last map is c_i, is exact.

Definition 7.8.6. The map c_i is called the **connecting homomorphism**, and the sequence (7.8.1) is called the **long exact sequence of cohomology**.

Problem 7.8.7. Let $C_\bullet$ and $D_\bullet$ be complexes of modules over a commutative ring A. Define the tensor product complex $(C \otimes D)_\bullet$ by the formula

$$(C \otimes D)_i = \bigoplus_{j+m=i} C_j \otimes_A D_m,$$

with differentials

$$d_i^{C\otimes D}|_{C_j \otimes D_m} = d_j^C \otimes 1 + (-1)^j \cdot 1 \otimes d_m^D.$$